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Ecological assessment of *Channa gachua* (Hamilton, 1822) in relation to physico-chemical parameters of Kosi River, Madhepura, Bihar

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Abstract- This study investigates the seasonal abundance and ecological associations of *Channa gachua* (Hamilton, 1822) in relation to key physicochemical parameters of Kosi River, a transboundary river flows through Madhepura district, Bihar. From September 2021 to August 2022, five fixed stations were monitored bimonthly to record water quality parameters-temperature, pH, dissolved oxygen (DO), free carbon dioxide (CO_2), total alkalinity, total hardness, nitrate, and phosphate-alongside fish population data. *C. gachua* was present throughout the year but exhibited peak abundance during pre-monsoon and post-monsoon seasons, coinciding with favorable temperature (28-31°C) and DO levels (>5.5 mg/L) conducive to breeding. A strong positive correlation (r = +0.68) was observed between DO and fish density, whereas negative correlations were noted with free CO_2 (r = -0.54) and turbidity. The species showed a preference for shallow, vegetated zones with muddy substrates, often dominated by Hydrilla and Nymphaea. The study underscores the role of *C. gachua* as a bio-indicator of wetland health and highlights its ecological resilience, making it a potential candidate for low-input aquaculture. Given the growing anthropogenic pressures on floodplain wetlands, periodic ecological assessments and habitat conservation strategies are recommended to sustain native fish diversity and wetland ecosystem integrity in North Bihar.

Key words: *Channa gachua,* seasonal abundance, water quality, dissolved oxygen, floodplain wetland, bio-indicator

INTRODUCTION

Channa gachua (Hamilton, 1822), commonly referred to as the dwarf snakehead, is a small, air-breathing freshwater fish recognized for its significant ecological role and nutritional value. This species is extensively distributed across the lentic (still water) and semi-lotic (slow-moving water) ecosystems of South Asia, particularly thriving in shallow wetlands, floodplains, and seasonal ponds.[1,2] Its notable adaptability includes a high tolerance for low dissolved oxygen levels and the ability to survive in adverse environmental conditions, such as desiccation, pollution, and periodic habitat drying. This resilience, combined with its opportunistic breeding behavior, allows *C. gachua* to persist in habitats that are typically inhospitable to other aquatic species.[3,4]

Kosi River a transboundary river flows through Madhepura district in Bihar, is primarily replenished through monsoon flooding. It retains water for most of the year and acts as a vital aquatic habitat, supporting a diverse range of aquatic vegetation and faunal communities.[5] However, the ecological balance of this river is increasingly threatened by human-induced disturbances, including land encroachment, indiscriminate disposal of domestic waste, and expanding agricultural activities in its vicinity.

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This study is designed to investigate the habitat preferences of *Channa gachua* within this dynamic freshwater system. It seeks to monitor the seasonal fluctuations in its population abundance and analyze how these changes relate to key physicochemical parameters of the water-such as temperature, dissolved oxygen, pH, turbidity, and nutrient load. The findings aim to provide insights into the environmental requirements and ecological resilience of *C. gachua*, contributing to the development of informed conservation and management strategies for river dependent fish species.

MATERIALS & METHODS

The present study was conducted over a period of one year, from September 2021 to August 2022, at five predetermined sampling stations of Kosi River near Madhepura. These sites were selected to represent the spatial variability of the wetland. Bimonthly surveys (once every two months) were carried out to capture seasonal trends and ecological fluctuations in water quality and fish abundance.

Water Quality Assessment

A comprehensive set of physicochemical parameters was analyzed to assess the ecological condition of the wetland. The parameters included- Water temperature, pH (potential of hydrogen), Dissolved Oxygen (DO), estimated using the Winkler titration method, Free carbon dioxide (CO_2), Total alkalinity, Total hardness, Nitrate (NO_3^-), Phosphate (PO_4^{3-})

All parameters were measured following standard protocols as prescribed by the American Public Health Association (APHA, 2012)[6], ensuring accuracy and consistency in data collection.

Fish Sampling and Enumeration

Fish sampling focused specifically on *Channa gachua* populations and was conducted during the early morning hours (between 6:00 AM and 9:00 AM), when fish activity is typically higher and capture is more efficient. A combination of traditional and effective sampling techniques was employed, including:

- Fine-meshed hand nets for capturing smaller individuals
- Bamboo fish traps for passive sampling
- Cast nets for broader coverage in open water areas

All captured specimens were handled carefully. Live individuals of *C. gachua* were counted and released back into the habitat after recording essential biometric data such as total length and body weight using calibrated measuring scales.

To quantify relative abundance, the Catch Per Unit Effort (CPUE) was calculated and expressed as the number of individuals caught per hour of active sampling. This metric enabled comparisons across sites and seasons, providing insight into population dynamics and habitat utilization by *C. gachua*.

RESULT & DISCUSSION

Physico-Chemical Characteristics of Water

The analysis of key physicochemical parameters across the study period (September 2021 - August 2022) revealed moderate seasonal variations, all remaining within permissible limits for sustaining freshwater fish life, including *Channa gachua*. The findings are summarized in the table 1.

Table 1- Physicochemical parameters across the study period with respect to seasons

Parameter	Range Recorded	Seasonal Trends
Water Temperature (°C)	21.6 – 33.5	Highest in May (summer), lowest in January (winter)
pH	6.5 – 8.2	Ranged from neutral to slightly alkaline, indicating a healthy aquatic environment
Dissolved Oxygen (DO)	4.1 – 7.6 mg/L	Relatively higher during winter and post-monsoon, correlating with reduced microbial oxygen demand and lower temperatures
Free Carbon Dioxide (CO_2)	2.6 – 5.8 mg/L	Peak levels observed during the monsoon season, likely due to increased organic decomposition and runoff
Total Alkalinity	114 – 132 mg/L	Fairly stable year-round, with a slight rise during the pre-monsoon period
Total Hardness	66 – 81 mg/L	Displayed minimal seasonal variation, indicating stable mineral content in the water
Nitrate (NO_3^-)	1.4 – 2.8 mg/L	Values suggest moderate eutrophication, influenced by nutrient runoff from nearby agricultural lands
Phosphate (PO_4^{3-})	0.8 – 2.5 mg/L	Slight increase recorded during monsoon, possibly due to soil leaching and external nutrient input

Overall, these parameters provided a suitable environment for the survival and reproduction of *C. gachua*, with no extreme fluctuations observed.

Seasonal variation in the abundance of *Channa gachua*

The abundance of *C. gachua* showed clear seasonal patterns, expressed as average catch per unit effort (CPUE in number/hour).

Table 2- The seasonal Catch Per Unit Effort (CPUE)

Season	Average CPUE (no./hr)	Remarks
Pre-monsoon	16.2	Marked by peak reproductive activity, increased catch rates, and high adult density
Monsoon	12.7	Slight decline due to habitat expansion and water dispersion caused by flooding
Post-monsoon	15.4	Increase in juvenile population, indicating successful breeding and recruitment
Winter	10.3	Decline in activity due to lower temperatures, with individuals seeking refuge in deeper, vegetated zones

Highest CPUE values were recorded in April (pre-monsoon) and November (post-monsoon), aligning with peak breeding and post-spawning phases. *C. gachua* showed a strong preference for shallow, vegetated margins with a muddy bottom and abundant submerged macrophytes like Hydrilla and Nymphaea, which offer both shelter and feeding grounds. These results highlight the ecological adaptability and seasonal dynamics of *Channa gachua* within a semi-natural wetland system under moderate anthropogenic stress.

Channa gachua exhibited distinct seasonal preferences, with noticeably higher abundance recorded during the pre-monsoon and post-monsoon periods. These seasons offered favorable environmental conditions, particularly moderate dissolved oxygen (DO) levels and optimal water temperatures ranging between 28°C and 31°C, which are known to support active breeding and metabolic processes. The fish population declined noticeably during the winter months, likely due to reduced metabolic activity associated with colder temperatures and limited habitat availability caused by water level reduction and contraction of shallow zones.

Statistical analysis revealed a strong positive correlation between dissolved oxygen and *C. gachua* density (r = +0.68), indicating that higher DO levels support greater population abundance. In contrast, negative correlations were observed with free carbon dioxide (r = -0.54) and turbidity, suggesting that elevated CO_2 levels and decreased water clarity negatively influence the species' distribution, possibly by impairing respiration or foraging efficiency.

Although *C. gachua* possesses a facultative air-breathing mechanism that enables it to endure hypoxic (low-oxygen) conditions, the data indicates that its optimal abundance was still associated with moderate DO levels (above 5.5 mg/L), where its physiological and reproductive performance is likely enhanced.

These results are consistent with earlier studies. For instance, Gupta *et al*. (2018)[7] documented comparable seasonal fluctuations and environmental preferences in *C. gachua* populations inhabiting the floodplain wetlands of Assam. Similarly, Singh and Pathak (2020)[8] reported that the species showed a marked affinity for shallow, vegetation-rich microhabitats, particularly those enriched with organic detritus, which provide both shelter and ample food resources.

Overall, the findings underscore the importance of maintaining moderate oxygen regimes and structurally complex habitats for sustaining healthy populations of *Channa gachua*, especially in wetlands under anthropogenic pressure.

CONCLUSION

The present study highlights the ecological resilience and environmental sensitivity of *Channa gachua* in a natural freshwater ecosystem. Its consistent occurrence across seasons, with peak abundance during pre- and post-monsoon periods, reflects its adaptability to moderate fluctuations in water quality and hydrological regimes. The strong positive association with dissolved oxygen and preference for shallow, vegetation-rich habitats underscore the species' ecological requirements and make it a reliable bio-indicator of wetland health.

Despite its tolerance to hypoxic conditions through facultative air-breathing, *C. gachua* thrives best under moderately oxygenated, freshwater conditions. This ecological trait, along with its reproductive plasticity, renders it suitable for community-based aquaculture, especially in low-input rural systems across eastern India.

However, the increasing anthropogenic pressures on Kosi River -such as nutrient runoff, habitat encroachment,

and waste disposal-pose significant risks to the sustainability of its native fish populations. Conservation strategies should prioritize periodic monitoring, protection of littoral vegetation, and control of pollution sources to preserve the ecological integrity of the river.

In conclusion, *Channa gachua* not only serves as a key component of the local aquatic biodiversity but also offers valuable insights for wetland management, sustainable fisheries development, and biodiversity conservation in floodplain habitats of North Bihar.

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